TASK 05-50-03-200-801-A
EFFECTIVITY: ALL

1. Hard Landing and Off-Temperature Envelope - Inspection/Check

A. General

(1) After an unusual landing condition, when a hard landing can occur, it is necessary to do inspections to make sure that the aircraft integrity is intact.

(2) An unusual landing condition is considered when the loads are out of the design envelope. This envelope includes a combination of the aircraft weight, vertical CG acceleration, roll rate, and external temperature at touchdown.

(3) This task includes:

(a) Hard Landing, which is when the vertical CG acceleration, roll, or pitch rate are more than their thresholds at touchdown.

(b) Off-temperature landing, which is when the temperature envelope is more than its threshold at touchdown.

NOTE: The inspection after an off-temperature landing is only necessary when the aircraft lands below the temperature envelope (below -45 °C).

(4) If an overweight landing occurs, refer to AMM TASK 05-50-09-210-801-A/600.

(5) A hard landing inspection is necessary when the crew members report a possible hard landing based on their judgment, or when the analysis of the flight operation data shows that the envelope of the landing parameters was not obeyed.

(6) The flight data source can be either DVDR 1, DVDR 2 or TCRF. Any of them is sufficient to define if a hard landing occurred and procedure with proper inspections.

NOTE: DVDR 1 is installed in the forward electronic compartment and DVDR 2 is installed in the aft electronic compartment.

(7) The operator can do the analysis of the flight operation data, but there is the alternative to send the data for an EMBRAER analysis.

(8) These maintenance technicians are necessary to do this task:

- 2 - Airframe (Structure)

B. References

REFERENCE	*DESIGNATION*
AMM MPP 32-11-01/400	MAIN-LANDING-GEAR SHOCK STRUT - REMOVAL/ INSTALLATION
AMM MPP 32-11-13/400	MAIN-LANDING-GEAR LOCKING STAY - REMOVAL/ INSTALLATION
AMM MPP 54-50-21/400	PYLON FORWARD FAIRINGS - REMOVAL/INSTALLA-TION
AMM MPP 54-50-25/400	PYLON AFT ACCESS PANELS - REMOVAL/INSTAL-LATION

(Continued)

REFERENCE	*DESIGNATION*
AMM SDS 32-12-00/1	MAIN-LANDING-GEAR DOORS
AMM SDS 32-21-00/1	NOSE-LANDING-GEAR
AMM SDS 52-11-00/1	FORWARD PASSENGER DOOR
AMM SDS 52-12-00/1	AFT PASSENGER DOOR
AMM SDS 52-31-00/1	FORWARD CARGO DOOR
AMM SDS 52-32-00/1	AFT CARGO DOOR
AMM SDS 52-41-00/1	FORWARD SERVICE DOOR
AMM SDS 52-42-00/1	AFT SERVICE DOOR
AMM TASK 05-50-09-210-801-A/600	Overweight Landing - Inspection/Check
AMM TASK 06-41-00-800-801-A/100	Fuselage Access Doors and Panels
AMM TASK 07-10-01-500-801-A/200	Complete Aircraft Jacking - Maintenance Practices
AMM TASK 20-00-00-910-801-A/200	Aircraft Safety Procedures Before Maintenance - Standard Procedures
AMM TASK 20-00-00-910-802-A/200	Aircraft Safety Procedures After Maintenance - Standard Procedures
AMM TASK 24-23-00-710-801-A/500	Ram-Air-Turbine (RAT) Manual Deployment - Operational Test
AMM TASK 32-11-07-000-801-A/400	Main-Landing-Gear Wheel Axle - Removal
AMM TASK 32-11-07-400-801-A/400	Main-Landing-Gear Wheel Axle - Installation
AMM TASK 32-11-11-000-801-A/400	Main-Landing-Gear Side Stay - Removal
AMM TASK 32-21-03-000-801-A/400	Nose-Landing-Gear Drag Brace - Removal
AMM TASK 32-21-05-000-801-A/400	Nose-Landing-Gear Locking Stay - Removal
AMM TASK 32-21-12-000-801-A/400	Nose-Landing-Gear Wheel Axle - Removal
AMM TASK 32-49-11-000-801-A/400	Brake Assembly - Removal
AMM TASK 32-49-11-400-801-A/400	Brake Assembly - Installation
AMM TASK 32-61-07-820-801-A/200	Main-Landing-Gear Downlock Sensor (Clearance) - Adjustment
AMM TASK 32-62-03-820-801-A/200	Main-Landing-Gear Weight-Off-Wheels Sensor (Clearance) - Adjustment
AMM TASK 51-50-00-820-801-A/200	Aircraft Alignment Check
AMM TASK 52-48-00-010-801-A/400	APU Doors - Opening
AMM TASK 53-04-20-000-801-A/400	Wing-to-Fuselage Fairing FWD Panel I - Removal
AMM TASK 53-04-20-400-801-A/400	Wing-to-Fuselage Fairing FWD Panel I - Installation
AMM TASK 53-04-40-000-801-A/400	Wing-to-Fuselage Fairing LH Side Panel - Removal
AMM TASK 53-04-40-400-801-A/400	Wing-to-Fuselage Fairing LH Side Panel - Installation
AMM TASK 53-04-41-000-801-A/400	Wing-to-Fuselage Fairing RH Side Panel - Removal
AMM TASK 53-04-41-400-801-A/400	Wing-to-Fuselage Fairing RH Side Panel - Installation
AMM TASK 53-04-50-000-801-A/400	Wing-to-Fuselage Fairing Aft Panel - Removal
AMM TASK 53-04-50-400-801-A/400	Wing-to-Fuselage Fairing Aft Panel - Installation

(Continued)

REFERENCE	*DESIGNATION*
NDT PART 5 TASK 57-51-00-213-801-B	Inboard Flap Torque Tube and Flap Mechanism Structures - Internal Detailed Visual Inspection
NDT PART 5 TASK 57-51-00-213-802-B	Inboard and Outboard Flap Mechanism Structures - Internal Detailed Visual Inspection
PW1900G PPMM PW1000G-A-72-00-00-09C-0B5A-A	Do An Inspection of the Engine After High Turbulence or Lateral Loads; Heavy or Overweight Landings

C. Zones and Accesses

ZONES ACCESS	*LOCATION*	*REFERENCE*
124BR	RH FWD Fuselage Bottom - Front Area	AMM TASK 06-41-00-800-801-A/100
125AL	LH FWD Fuselage Bottom - Rear Area	AMM TASK 06-41-00-800-801-A/100
191AR	Wing-to-Fuselage Fairing FWD Panel I	AMM TASK 06-41-00-800-801-A/100
191BL	Wing-to-Fuselage Fairing FWD Panel I	AMM TASK 06-41-00-800-801-A/100
191CL	Wing-to-Fuselage Fairing FWD Panel I	AMM TASK 06-41-00-800-801-A/100
191EL	Wing-to-Fuselage Fairing FWD Panel I	AMM TASK 06-41-00-800-801-A/100
192AL	Wing-to-Fuselage Fairing LH Center Panel	AMM TASK 06-41-00-800-801-A/100
192BL	Wing-to-Fuselage Fairing LH Center Panel	AMM TASK 06-41-00-800-801-A/100
193BR	Wing-to-Fuselage Fairing Center Panel	AMM TASK 06-41-00-800-801-A/100
193CR	Wing-to-Fuselage Fairing Center Panel	AMM TASK 06-41-00-800-801-A/100
198AR	Wing-to-Fuselage Fairing RH Aft Panel	AMM TASK 06-41-00-800-801-A/100
198BR	Wing-to-Fuselage Fairing RH Aft Panel	AMM TASK 06-41-00-800-801-A/100
198DR	Wing-to-Fuselage Fairing RH Aft Panel	AMM TASK 06-41-00-800-801-A/100
198HR	Wing-to-Fuselage Fairing RH Aft Panel	AMM TASK 06-41-00-800-801-A/100
351CL	Tail Cone	AMM TASK 06-41-00-800-801-A/100
351CR	Tail Cone	AMM TASK 06-41-00-800-801-A/100
351GL	Tail Cone	AMM TASK 06-41-00-800-801-A/100

D. Tools and Equipment

REFERENCE	*DESIGNATION*
Commercially available	Flashlight

E. Hard-Landing Detection Procedure

SUBTASK 210-002-A

(1) Figure 601, Figure 602, and Figure 603 show a flowchart that is a summary of the necessary actions when the crew or ACMF, TCRF, reports a possible hard landing, when the hard landing is detected by flight data analysis or when the aircraft landed out of the temperature envelope.

(2) When the crew or ACMF or TCRF reports a possible hard landing, do MLG phase I and NLG phase I inspections immediately. If no discrepancy is found during phase I inspection, a fly-by of 10 FC is permitted. During the fly-by period, do one of these two procedures:

NOTE: The ACMF or TCRF reports or the absence of these do not replace or have any priority over flight crew reports.

(a) Do phase II inspection. In this case, no flight data analysis is necessary. Do phase III inspection only if you find any discrepancy during phase II inspection.

NOTE:
- Make sure that phase I inspection is done before starting phase II.
- If phase III inspection is necessary, make sure that you did all phase II inspections.
- Once the flight data were not analyzed, you do not know which landing gear had a hard landing. Thus, do MLG phase II and NLG phase II inspections.

(b) Obey the instructions in this section to make an analysis the flight data. Do phase II inspection only if the data analysis shows that the landing conditions were out of the threshold limits given in this section. Do phase III inspection only if you find any discrepancy during phase II inspection.

NOTE:
- Make sure that phase I inspection is done before starting phase II.
- If phase III inspection is necessary, make sure that you did all phase II inspections.
- The flight data download must be done within a maximum of 20 FH. If not, the data can be missed because of the recording time of the unit.

(3) When the source for you to find a hard-landing condition is the flight data analysis, do the required inspections according to Figure 602. Do phase III inspection only if you find any discrepancy during phase II inspection.

NOTE:
- Make sure that phase I inspection is done before starting phase II.
- If phase III inspection is necessary, make sure you did all phase II inspections.
- If it is not possible to do phase II inspection immediately, a fly-by of 10 FC is permitted from the moment of the hard landing is found through the flight data analysis, provided that phase I inspection is done immediately and no damage is found during this inspection.
- If any discrepancy is found during phase I inspection, phase II inspection must be done immediately.
- During the fly-by period, phase II inspection must be done.
- Once the landing gear that was out of the envelope is known, do the inspections related to this landing gear only.

(4) When the crew members or ACMF or TCRF report a possible hard landing, make the analysis for the MLG and NLG.

(5) Make the analysis of the hard landing separately from the MLG and NLG. The MLG hard touchdown is a result of a high vertical CG acceleration or high roll rates at the MLG touchdown. The NLG hard touchdown can occur because of high pitch rates when the NLG touches the runway.

(6) Once the hard landing analysis is made separately for the MLG and NLG, the necessary inspections depend on which landing gear had a hard landing.

(7) Use the parameters below to make an analysis of the flight data to find a hard landing:

- Air/Ground indication
- Gross weight
- Normal CG acceleration (recorded in Normal (Vertical) Acceleration (DVDR) or fdrAccelNormal-1a (TCRF))
- Pitch rate (recorded in AHRS Inertial Pitch Rate (DVDR)
- Pitch angle (AHRSPitchAng (DVDR) pitchAngle-1a (TCRF))
- Roll rate (recorded in AHRS Inertial Roll Rate (DVDR) or rollAttRate-1a (TCRF))
- Time (GMT)
- Date (Day)
- Flight number (FltNumber)
- Nose Landing Gear WOFW Sensor 1" (DVDR) or all Gear WOW2 (TCRF)

NOTE: To find the sampling rate of the parameters recorded in TCRF, refer to 196INO001. Similarly, to find the sampling rate of the parameters recorded in DVDR, refer to 196EBD717. Both reports are available for download at Embraer website (Flyembraer.com > Support Center > download_center > Commercial Jets > Maintenance > Technical Support > E2 > Aircraft Software and Reports): TCRF and FDR."

(8) If a landing was done out of the temperature envelope (below -45 °C), the MLG and NLG hard landing detection procedures are not applicable. Then, do as follows:

(a) If the landing was done below -45 °C, do phase II inspection. Do phase III inspection only if you find a discrepancy during phase II inspection.

NOTE: If phase III inspection is necessary, make sure that you did all phase II inspections.

(9) The procedures given herein let you find a hard-landing condition, find the required inspections, and put the aircraft back to a serviceable condition.

(10) If you will not do phase III inspection, it is not necessary to send data to Embraer. But, if you will, send the flight data and report of the damage found to Embraer for analysis.

(11) In situations when the aircraft touches the ground with the NLG first, the MLG and NLG hard landing detection procedures with flight data are not applicable and phase II inspections for MLG and NLG must be done.

(12) When the conditional inspection shows that you must examine a component, look for the conditions that follow:

- Pulled-apart structure
- Loose paint (paint flakes)
- Discoloration
- Distortion (twisted parts)
- Wrinkles or buckles in the structure
- Bent components
- Fastener holes that became oversized
- Loose fasteners
- Fasteners that pulled out or are missing
- Delamination
- Fiber breakouts
- Black signs around fasteners in composites ("smoking" fasteners)
- Misalignment
- Interference
- Nicks or gouges
- Scratches
- Other signs of damage.

(13) All the inspections are classified as GVI.

F. MLG Hard-Landing Detection

SUBTASK 210-003-A

(1) The MLG hard landing detection is done through the check of the vertical CG acceleration and roll rate check. When this parameter is out of its thresholds, the landing is a hard landing.

(2) Vertical CG Acceleration Check

(a) In the downloaded flight data, identify the initial and final data samples to specify the range for the vertical CG acceleration analysis (Figure 604):

1 The range starts 3 s before the first "Ground" recorded for the air/ground indication (Air/Gnd).

2 The range stops at the first pitch angle (Pitch) sample with a value less than or equal to -0.2 degrees after the transition of the Air/Ground indication (Air/Gnd) from "Air" to "Ground".

(b) Get the maximum value of the vertical CG acceleration (NormAccel) in the selected range.

NOTE:
- If the maximum value is between X.YZ00 and X.YZ49, it must be rounded down to X.(YZ);
- If the maximum value is between X.YZ50 and X.YZ99, it must be rounded up to X.(YZ+1).

(3) *EFFECTIVITY: EMBRAER 190-E2 ACFT*

Get the maximum value of the roll rate and do the applicable action according to Figure 605 or Figure 606 .

(4) *EFFECTIVITY: EMBRAER 195-E2 ACFT*

Get the maximum value of the roll rate and do the applicable action according to Figure 607 , Figure 608 or Figure 609 .

G. NLG Hard-Landing Detection

SUBTASK 210-009-A

(1) In the Flight Data, the hard landing acquisition is based on first sample with pitch angle lower than 4 degrees and 1 second after the Nose landing Gear have weight on wheels.

As some versions of the TCRF does not record the Nose Landing gear WOW, the ALL gear WOW have to used for determine the nose landing gear is WOW, for the TCRF.

In this interval, the maximum absolute result of the pitch rate and must be considered for hard landing analysis (Figure 604).

(2) *EFFECTIVITY: EMBRAER 190-E2 ACFT*

If the maximum pitch rate is greater than or equal to the threshold of -10.0 degrees per second, no other action is necessary.

(3) *EFFECTIVITY: EMBRAER 195-E2 ACFT*

If the maximum pitch rate is greater than or equal to the threshold of -8.0 degrees per second, no other action is necessary.

(4) If the maximum pitch rate is less than the mentioned threshold, do NLG phase II inspection.

NOTE:
• Make sure that phase I inspection is done before starting phase II.
• If it is not possible to do phase II inspection immediately, a fly-by of 10 FC is permitted from the moment of the hard landing detection through the flight data analysis, provided that phase I inspection is done immediately and no damage is found during this inspection.
• If any discrepancy is found during phase I inspection, phase II inspection must be done immediately.
• During the fly-by period, phase II inspection must be done.

(5) If no discrepancy is found during NLG phase II inspection, no other action is necessary. If you find any discrepancy during NLG phase II inspection, do NLG phase III inspection and send Embraer this data:

(a) Flight Data Source data that contains the hard landing incident.

(b) Damage report with information about the damage found during the inspections.

H. Job Set-Up

SUBTASK 840-002-A

(1) Do the procedure to make the aircraft safe for maintenance (AMM TASK 20-00-00-910-801-A/200).

SUBTASK 940-002-A

(2) Set all flight control surfaces to the neutral position.

I. MLG - phase I Inspection

SUBTASK 210-010-A

(1) Visually examine these MLG items but do not disassemble them:

NOTE: It is necessary to disassemble the MLG if you think that there is damage related to a hard landing after an external inspection.

(a) Examine the MLG wheels for distortion, bending, dents, and fracture.

(b) Examine the MLG tires for general condition and deflation.

(c) Examine the brake assembly for damage, and the hoses for signs of fluid leakage.

(d) Examine the MLG wheelwell for signs of fluid leakage.

(e) Examine the MLG locking stay for distortion, bending, and fracture.

(f) Examine the upper and lower MLG side stays for distortion, bending, and fracture.

(g) Examine the MLG shock absorber conditions.

(h) Examine the MLG trailing arm conditions.

(2) Fuselage

(a) Examine the external surface of the fuselage for fluid leakage.

(b) Externally examine the areas near the wing stub for signs of damage.

(c) Open and close the aft passenger door and make sure that it operates correctly (AMM SDS 52-12-00/1).

(d) Open and close the aft service door and make sure that it operates correctly (AMM SDS 52-42-00/1).

(e) Open and close the forward and aft cargo doors and make sure that they operate correctly (AMM SDS 52-31-00/1 and AMM SDS 52-32-00/1).

(f) Open and close the main landing gear inboard door and make sure that it operates correctly (AMM SDS 32-12-00/1).

(g) Externally examine the aft cargo door metallic and composite scuff-plate for signs of bend.

(h) Examine the tail cone and silencer.

(i) Examine the wing-to-fuselage fairing junction with the fuselage and look for missing or damaged screws, missing sealant, and paint peel-off.

(3) Wing

(a) Examine the MLG doors for misalignment and signs of damage.

(b) Examine the external surface of the wing for fuel or other fluid leakage.

NOTE: Carefully examine the fuel-tank access panel contours for fuel leakage.

(c) Examine the pylon fairings for signs of damage.

(4) Flight Controls

(a) With the flaps, slats and spoilers fully retracted and ailerons, rudder and elevators in the neutral position, examine all surfaces for misalignment.

(b) With the flight controls fully deflected, examine the internal mechanisms for bent components.

(c) Examine all flight controls to make sure that they move freely.

NOTE: If necessary, use a flashlight to do the inspection of the inner side of the flap rod fairing for bent components.

(5) Engine

(a) Examine the inlet-cowl internal skin for signs of chafing against the fan blades.

J. MLG - phase II Inspection

SUBTASK 210-004-A

(1) Main Landing Gear

(a) Do an inspection of the lower skin of the rear fuselage around the jacking point before you put the aircraft on jacks. If there are signs of buckling, do not put the aircraft on jacks and speak to Embraer for more instructions.

(b) Put the aircraft on jacks (AMM TASK 07-10-01-500-801-A/200).

(c) Do a check of the clearances of the proximity sensors. The sensor surfaces must be parallel.

1 For the downlock sensor, refer to AMM TASK 32-61-07-820-801-A/200.

2 For the weight-on-wheels sensor, refer to AMM TASK 32-62-03-820-801-A/200.

(d) Remove the MLG side-stay lower bolt (AMM TASK 32-11-11-000-801-A/400) and examine for distortion, bending, and fracture.

(e) Examine the side-stay lower bolt lug for distortion, bending, and fracture. Refer to Figure 612, Sheet 1.

(f) Examine the trunnion fitting. Refer to Figure 612, Sheet 2.

(g) Retract the landing gear and do the visual check of the retraction path for interferences, clearances, misalignments, malfunctions and unusual noise in the landing gear, landing gear bay and landing gear door mechanisms. Extend the landing gear and do the check for unusual noise, uplock and downlock mechanisms malfunction. Do the check for any unusual behavior.

NOTE: An irregular, non-smooth operation of the landing gear can show deformation of the shock absorber and rear pintle pin. They are the most critical parts when a hard landing event occurs. Do the check of their condition and correct alignment during MLG phase III inspection.

(h) Do the retraction procedure again and extend the landing gear with the free-fall system. Do the check for any unusual behavior.

(i) Remove the brake assembly (AMM TASK 32-49-11-000-801-A/400).

(j) Examine the wheel axle to know if it is bent. Remove it, if necessary (AMM TASK 32-11-07-000-801-A/400).

(k) If removed, install the wheel axle (AMM TASK 32-11-07-400-801-A/400).

(l) Install the brake assembly (AMM TASK 32-49-11-400-801-A/400).

(m) In case of tire burst during the event, do the check of the side-stay condition.

(2) Fuselage

(a) Remove the forward panel of the wing-to-fuselage fairing (AMM TASK 53-04-20-000-801-A/400).

NOTE: As alternative step, you can remove only access panels 191AR, 191EL, 191BL, 191CL, and 191EL, and do a boroscope inspection.

(b) Remove access panels 192AL, 192BL, 193BR and 193CR (AMM TASK 06-41-00-800-801-A/100).

(c) Remove the side panels of the wing-to-fuselage fairing (AMM TASK 53-04-40-000-801-A/400, AMM TASK 53-04-41-000-801-A/400).

(d) Remove the aft panel of the wing-to-fuselage fairing (AMM TASK 53-04-50-000-801-A/400).

NOTE: As alternative step, you can remove only access panels 198AR, 198BR, 198DR, 198HR, and do a boroscope inspection.

(e) Externally examine the upper and lower fuselage skin panels for loose or sheared rivets, structural damage and signs of fuel or other fluid leakage. If you find wrinkles on the skin panels, do an internal visual inspection of the fuselage.

(f) Carefully examine all equipment in the wing-to-fuselage fairing region and its attachment to the fuselage.

(g) Externally examine the aft cargo door metallic and composite scuff-plate for signs of bend.

(h) In the forward electronic compartment, examine all equipment, racks, shelves and supports for integrity.

(i) In the middle electronic compartment, examine all equipment, racks, shelves and supports for integrity.

(j) In the rear fuselage, examine all equipment, racks, shelves and supports for integrity.

(k) Carefully examine the internal side of the fuselage on the attachment area of the galley to the frame. Refer to Figure 611, Sheet 1, Figure 611, Sheet 3 and/or Figure 611, Sheet 2.

NOTE: It is not necessary to remove the galley to get access to the inspection area.

(l) In the rear fuselage, examine the potable water tank and its attachment to the fuselage structure.

(m) In the rear fuselage, examine the vacuum waste-system tank and its attachment to the fuselage structure.

(n) Install the aft panel of the wing-to-fuselage fairing (AMM TASK 53-04-50-400-801-A/400).

(o) Install the side panels of the wing-to-fuselage fairing (AMM TASK 53-04-40-400-801-A/400, AMM TASK 53-04-41-400-801-A/400).

(p) Install access panels 192AL, 192BL, 193BR and 193CR (AMM TASK 06-41-00-800-801-A/100).

(q) Install the forward panel of the wing-to-fuselage fairing (AMM TASK 53-04-20-400-801-A/400).

(3) Wing

(a) Examine wing spars II and III. Start from the MLG wheelwell. Remove the access panels and, with the inboard flap in the down position, also examine the lower skin adjacent to them.

(b) Examine the upper and lower wing skin panels from the wing root to the wing tip.

(c) Externally examine the wing, nacelles, and engine struts for loose or sheared rivets, structural damage, and signs of fuel and other fluid leakage.

(d) Examine the wing-to-fuselage junction and MLG wheelwell for damage. Look for flaked paint and pulled-out or missing fasteners.

(4) Flight Controls

(a) Do an inspection of the flap rod and mechanism (NDT PART 5 TASK 57-51-00-213-801-B and NDT PART 5 TASK 57-51-00-213-802-B).

(b) With the flaps, slats and spoilers fully retracted and ailerons, rudder and elevators in the neutral position, examine all surfaces for misalignment.

(c) With the flight controls fully deflected, examine the internal mechanisms for bent components.

(d) Examine all flight controls to make sure that they move freely.

(5) Engine

(a) Do the engine overlimit condition inspection (Refer to PW1900G PPMM PW1000G-A-72-00-00-09C-0B5A-A).

(6) *EFFECTIVITY: ON ACFT WITH BIPARTITE TAIL CONE APU ACCESS DOOR*

APU

(a) Open the APU access doors (AMM TASK 52-48-00-010-801-A/400):

- 351CL
- 351CR.

(b) Examine the external side of APU for signs of damage.

(7) *EFFECTIVITY: ON ACFT WITH TAIL CONE APU ACCESS DOOR WITH HATCH*

APU

(a) Open the APU access door (AMM TASK 52-48-00-010-801-A/400):

- 351GL

(b) Examine the external side of APU for signs of damage.

K. MLG - phase III Inspection

SUBTASK 210-005-A

(1) Main Landing Gear

(a) Make sure that you did all phase II inspections.

(b) Do the inspection of the rear fuselage lower skin around the jacking point before you put the jack the aircraft. If there are signs of buckling, do not put the aircraft on jacks and contact Embraer for further instructions.

(c) Put the aircraft on jacks (AMM TASK 07-10-01-500-801-A/200).

(d) Remove the MLG locking stay pins (AMM MPP 32-11-13/400) and examine for distortion, bending, and fracture. Refer to Figure 612.

(e) Remove the MLG pintle pins AMM MPP 32-11-01/400 and examine for distortion, bending, and fracture. Refer to Figure 612.

(f) Install MLG serviceable locking stay pins (AMM MPP 32-11-13/400).

(g) Install MLG serviceable pintle pins (AMM MPP 32-11-01/400).

(h) Carefully examine the internal side of CF II in the area below the floor cross-beams and columns.

(i) Do the aircraft alignment check (AMM TASK 51-50-00-820-801-A/200).

(2) Pylon

(a) Remove the pylon rear access panels (AMM MPP 54-50-25/400).

(b) Remove the pylon forward fairing (AMM MPP 54-50-21/400).

(c) Do the inspection of the pylon structures that attaches to the wing. Refer to Figure 610 and do the inspection of the elements that follow:

Pylon Lugs

Links

Wing Lugs

(d) Install the pylon rear access panels (AMM MPP 54-50-25/400).

(e) Install the pylon forward fairing (AMM MPP 54-50-21/400).

(3) APU

(a) If any external damage is found at APU during phase I or II inspection, do the inspection as per APS2600E ENGINE MANUAL - PART NO. 4510729 -Task 05-50-00-214-804.

L. NLG - phase I Inspection

SUBTASK 210-006-A

(1) Visually examine these items but do not disassemble them:

NOTE: It is only necessary to disassemble the NLG if you think that there is damage related to the hard landing after an external examination.

(2) Nose Landing Gear

(a) Examine the NLG wheels for bending, fracture, dents, and wear.

(b) Examine the NLG tires for general condition and deflation.

(c) Examine the NLG strut for leakage around the sliding tube diameter.

(d) Examine the steering cylinder for signs of fluid leakage.

(e) Examine the NLG wheelwell for signs of fluid leakage.

(f) Examine the NLG locking stay for distortion, bending, and fracture.

(g) Examine the NLG torque links for distortion, bending, and fracture.

(h) Examine the NLG upper and lower drag braces for distortion, bending, and fracture.

(i) Examine the hydraulic actuators for bending, fracture of rod ends, and fluid leakage.

(3) Fuselage

(a) Examine the external surface of the fuselage for fluid leakage.

(b) Examine the NLG bay for signs of fluid leakage or distortion.

(c) Externally examine the lower fuselage skin panels.

NOTE: Carefully examine the rivets for integrity along the contour of the NLG bay.

(d) Examine the NLG doors and rods for distortion.

(e) Examine the NLG attachment fittings.

(f) Open and close the forward passenger door and make sure that it operates correctly (AMM SDS 52-11-00/1).

(g) Open and close the forward service door and make sure that it operates correctly (AMM SDS 52-41-00/1).

M. NLG - phase II Inspection

SUBTASK 210-007-A

(1) Nose Landing Gear

(a) Examine the lower skin of the rear fuselage around the jacking point before you put the aircraft on jacks. If there are signs of buckling, do not put the aircraft on jacks and speak to Embraer for more instructions.

(b) Put the aircraft on jacks (AMM TASK 07-10-01-500-801-A/200).

(c) Do a check of the clearances of the proximity sensors. The sensor surfaces must be parallel.

1 For the downlock sensor, refer to AMM TASK 32-61-07-820-801-A/200.

2 For the weight-on-wheels sensor, refer to AMM TASK 32-62-03-820-801-A/200.

(d) Examine the NLG attachment fittings.

(e) Retract the landing gear and do the visual check of the retraction path for interferences, clearances, misalignments, malfunctions and unusual noise in the landing gear, landing gear bay and landing gear door mechanisms. Extend the landing gear and do the check for unusual noise, uplock and downlock mechanisms malfunction. Do the check for any unusual behavior.

NOTE: An irregular, non-smooth operation of the landing gear can show deformation of the landing gear. You must then examine it for correct alignment.

(f) Do the retraction procedure again and extend the landing gear with the free-fall system. Do the check for any unusual behavior.

NOTE: An irregular, non-smooth operation of the landing gear can show deformation of the landing gear. You must then examine it for correct alignment.

(g) Examine the drag brace and locking stay link assemblies (AMM SDS 32-21-00/1).

(h) Examine the NLG wheelwell for buckled structure.

(i) Examine the NLG axle (AMM TASK 32-21-12-000-801-A/400) to see if it is cracked or bent.

(2) Fuselage

(a) Open these panels to get access to the internal side of the NLG wheelwell:

- 125AL
- 124BR

(b) Carefully examine the gray areas shown in Figure 614 for signs of damage or distortion.

(c) Examine the fuselage skin attachments above the landing gear beam for distortion. Look for flaked paint and pulled-out or missing fasteners.

(d) In the forward electronic compartment, examine all equipment, racks, shelves and supports for integrity.

(e) In the radome, do a check to see if the antennas are in position and examine the web structure for integrity.

(f) Do the operational test of the RAT deployment (AMM TASK 24-23-00-710-801-A/500).

(g) In the forward fuselage, examine the potable water tank and its attachment to the fuselage structure.

N. NLG phase III Inspection

SUBTASK 210-008-A

(1) Make sure you did all phase II inspections.

(2) Do the inspection of the rear fuselage lower skin around the jacking point before you put the aircraft on jacks. If there are signs of buckling, do not put the aircraft on jacks and speak to Embraer for further instructions.

(3) Put the aircraft on jacks (AMM TASK 07-10-01-500-801-A/200).

(4) Remove the NLG locking stay pins (AMM TASK 32-21-05-000-801-A/400) and examine for distortion, bending, and fracture. Refer to Figure 613.

(5) Remove the NLG upper and lower drag-brace pins (AMM TASK 32-21-03-000-801-A/400) and examine for distortion, bending, and fracture. Refer to Figure 613.

(6) Examine the landing gear for correct operation of the free-fall system.

NOTE: An irregular, not-smooth operation of the landing gear during free-fall can show deformation of the landing gear. You must then examine it for correct alignment.

(7) Do the aircraft alignment check (AMM TASK 51-50-00-820-801-A/200).

O. Job Close Up

SUBTASK 840-003-A

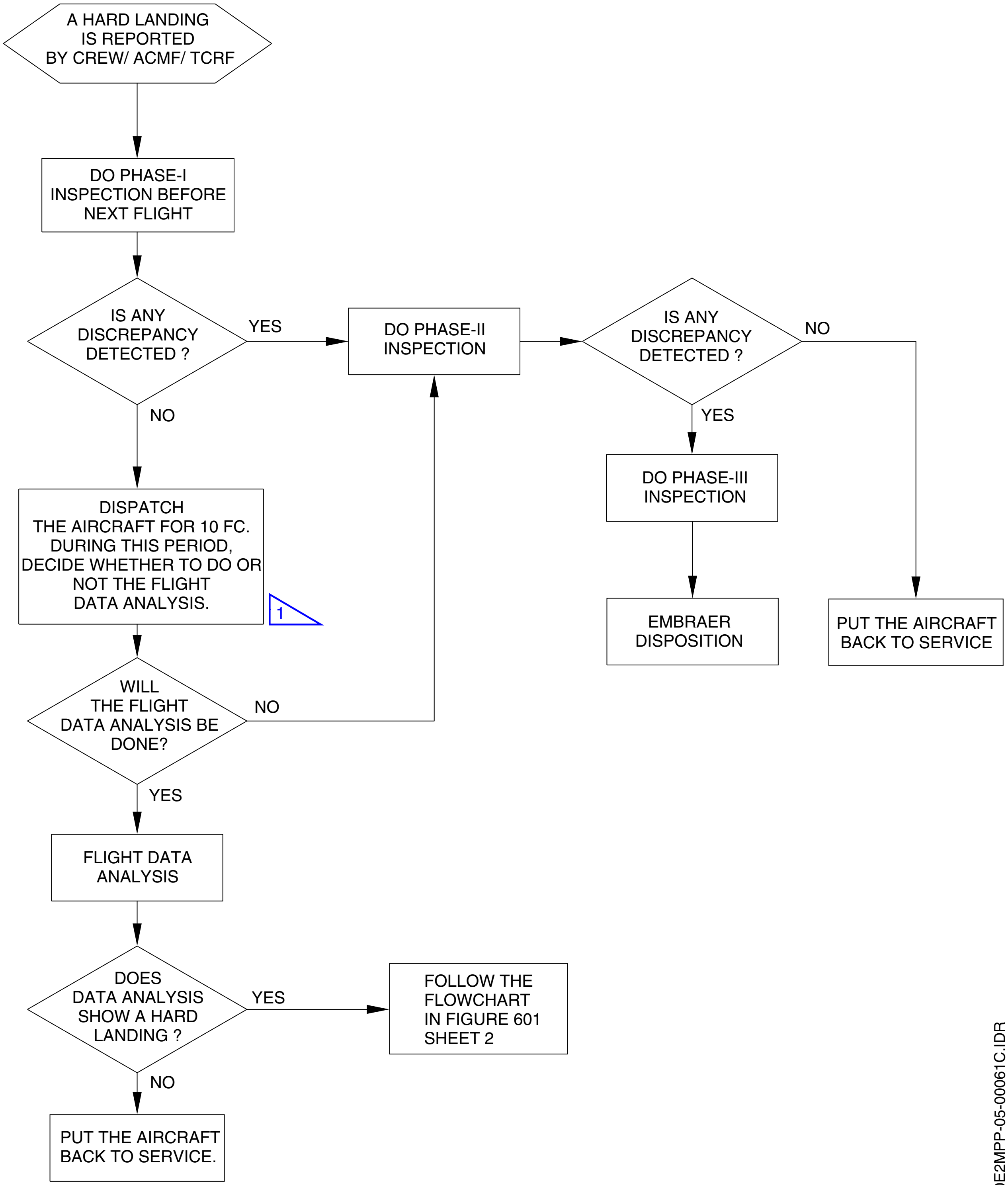
(1) Remove from the work area all tools, equipment, and materials that you used.

(2) Do the procedure to restore the aircraft after the maintenance procedures (AMM TASK 20-00-00-910-802-A/200).

EFFECTIVITY: ALL

Hard Landing Flowchart - Crew Report

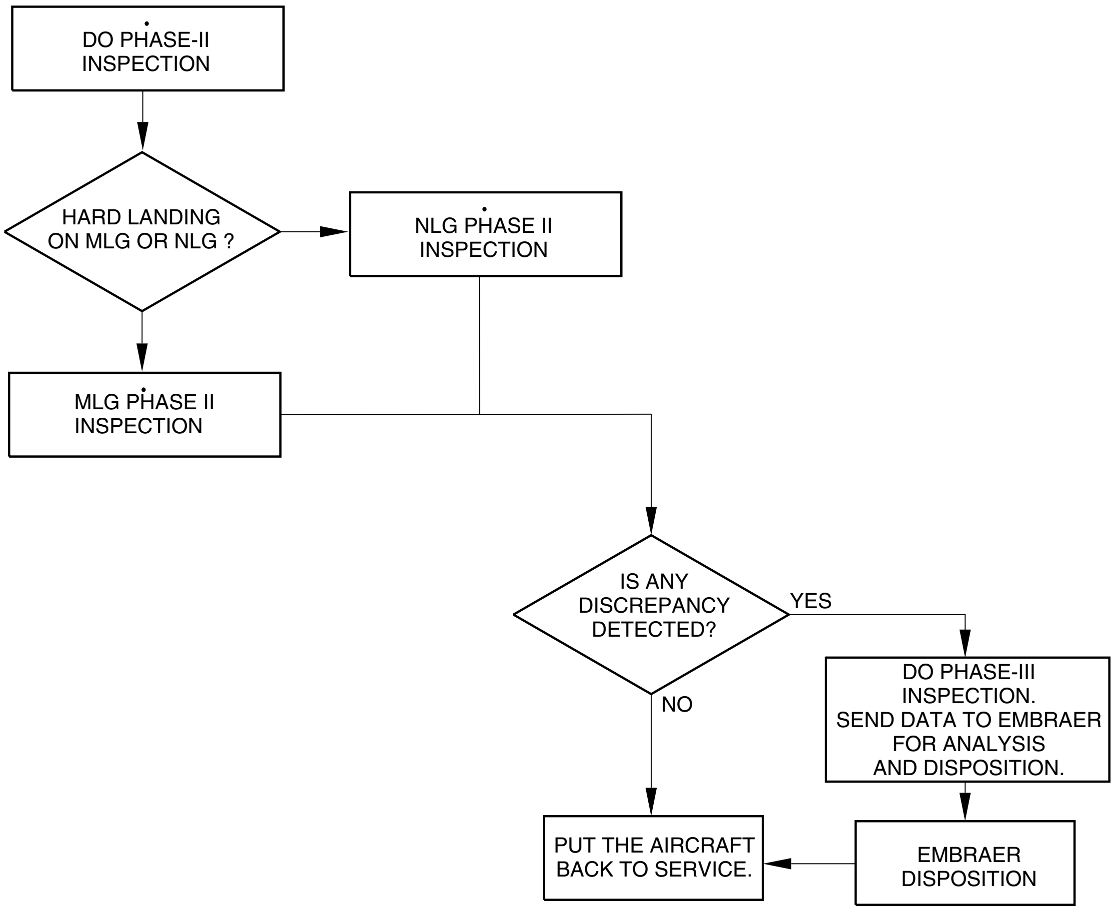
Figure 601 - Sheet 1



1 DO THE FLIGHT DATA DOWNLOAD IN A MAXIMUM OF 20 FH, OTHERWISE YOU CAN MISS THE DATA BECAUSE OF THE RECORDING TIME OF THE UNIT.

EM170E2MPP-05-00061C.IDR

EFFECTIVITY: ALL
Hard Landing Flowchart - Crew Report
Figure 601 - Sheet 2

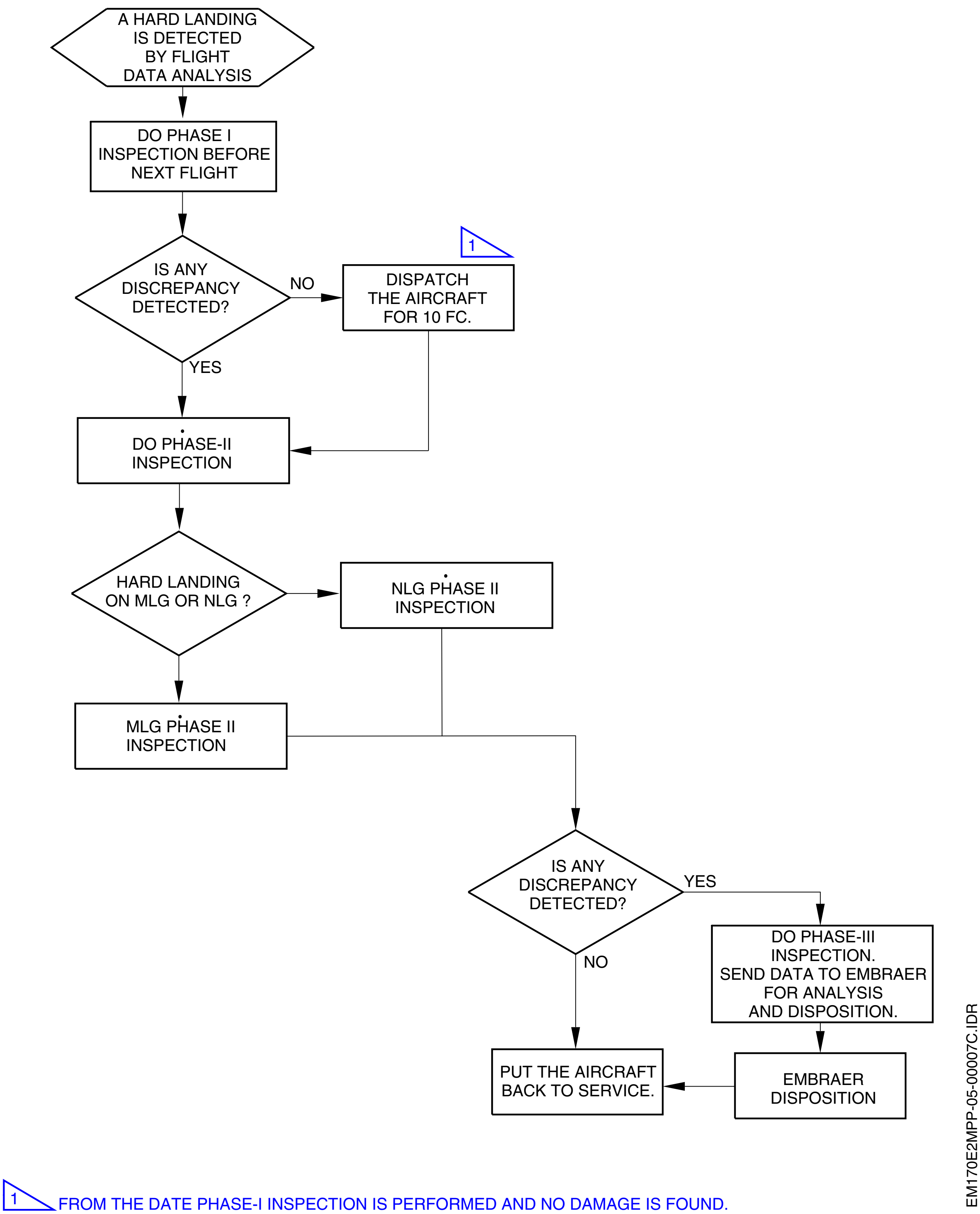


EM170E2MPP-05-00146A.IDR

EFFECTIVITY: ALL

Hard Landing Flowchart - Flight Data Analysis

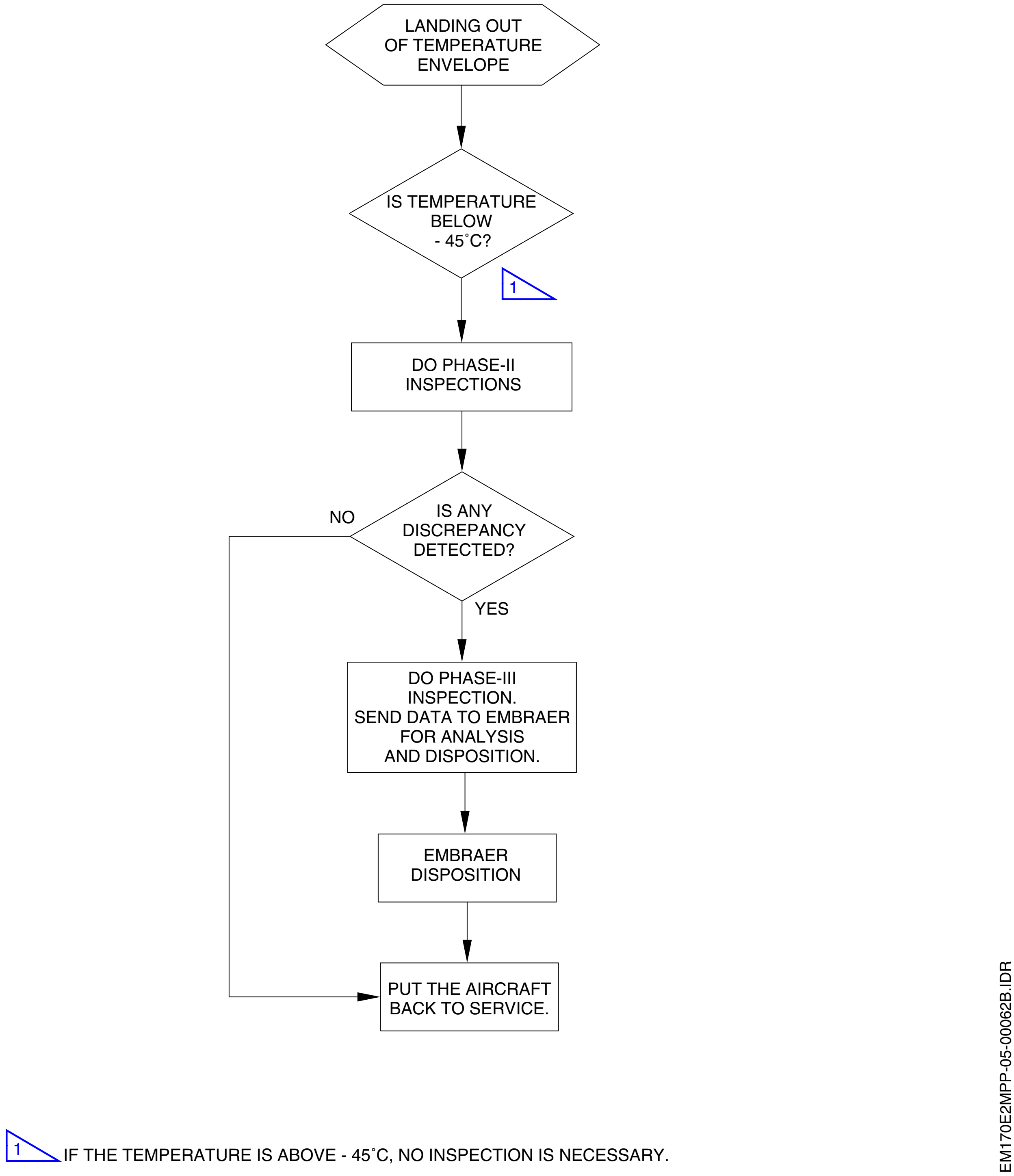
Figure 602



EFFECTIVITY: ALL

Hard Landing Flowchart - Out of Temperature Envelope

Figure 603



EFFECTIVITY: ALL

Downloaded DVDR Data

Figure 604

Sample	GMT	GrossWeight (lb)	Air/Gnd	Roll Rate (deg)	Pitch Angle (deg)	Pitch Rate (deg)	All Gear WOW2	NormAccel (g)
30443				2.3	6.2	-0.2	0	1.074
30443								1.07
30443					6.2	-0.3	0	1.066
30443								1.051
30444	22:00:05		AIR	2.6	6.3	-0.5	0	1.063
30444								1.051
30444					6.3	-0.7	0	1.051
30444								1.043
30444				2.5	6.5	-0.5	0	1.051
30444								1.051
30444					6.5	6.2	0	1.059
30444								1.066
30445	22:00:06	86600	AIR	1.6	6.7	6.2	0	1.059
30445								1.063
30445					6.9	6.3	0	1.047
30445								1.059
30445				0.5	6.9	6.3	0	1.059
30445								1.051
30445					7	6.5	0	1.063
30445								1.063
30446	22:00:07		AIR	-0.2	7.2	6.5	0	1.074
30446								1.074
30446					7.4	6.7	0	1.078
30446								1.07
30446				-0.4	7.4	6.9	0	1.074
30446								1.074
30446					7.6	6.9	0	1.07
30446								1.066
30447	22:00:08		AIR	-0.2	7.4	7	0	1.07
30447								1.07
30447					7	7.2	0	1.387
30447								1.527
30447				1.4	6.3	7.4	0	1.328
30447								1.258
30447					6.2	7.4	0	1.109
30447								0.906
30448	22:00:09		GROUND	2.5	4.9	7.6	0	0.953
30448								0.998
30448					4.4	7.4	0	0.863
30448								0.918
30448				1.4	3.9	7	0	1.09
30448								1.176
30448					2	6.3	0	1.199
30448								1.125
30449	22:00:10	86600	GROUND	0.9	1.3	6.2	1	0.996
30449								0.973
30449					-0.2	4.9	1	1.008
30449								1.023
30449				0.7	-0.3	4.4	1	1.043
30449								1.121
30449					-0.5	3.9	1	1.063
30449								0.922
30450	22:00:11		GROUND	0.7	-0.7	2	1	1.066
30450								1.074
30450					-0.5	1.3	1	1.074
30450								1.031

LEGEND:

A - (NORMAL ACCELERATION CHECK) - 3 SECONDS BEFORE THE FIRST "GROUND" RECORDED UP TO THE FIRST SAMPLE WITH PITCH ANGLE LOWER THAN OR EQUAL TO -0.2 DEGREE.

B - (ROLL RATE CHECK) - 3 SECONDS BEFORE THE FIRST "GROUND" RECORDED UP TO THE FIRST SAMPLE WITH PITCH ANGLE LOWER THAN OR EQUAL TO -0.2 DEGREE.

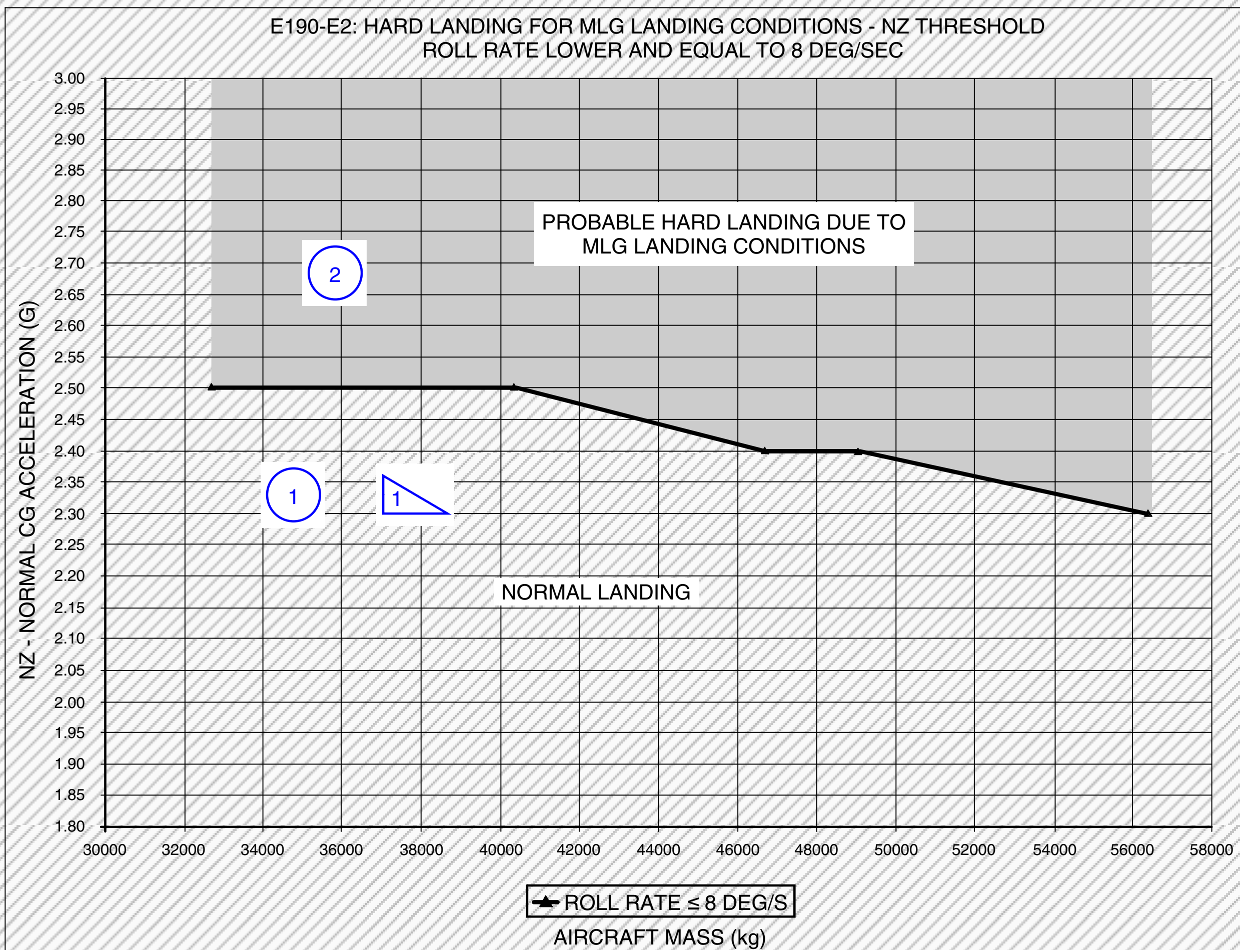
C - (PITCH RATE CHECK) - THE RANGE STARTS ON THE FIRST SAMPLE WITH PITCH ANGLE LOWER THAN 4 DEGRESS, THE RANGE STOPS 1 SECOND AFTER THE LAST SAMPLE WITH THE NLG IN AIR (DVDR) OR 1 SECOND AFTER THE LAST SAMPLE WITH ANY LG IN AIR (TCRF).

EM170E2MPP-05-00008D.IDR

EFFECTIVITY: EMBRAER 190-E2 ACFT

Roll rate <= 8 DEG/S

Figure 605



	NORMAL ACCELERATION (NZ) THRESHOLD (G)
AIRCRAFT MASS (kg)	ROLL RATE = 8 DEG/S
32700	2.50
37200	2.50
40300	2.50
46700	2.40
49050	2.40
56400	2.30

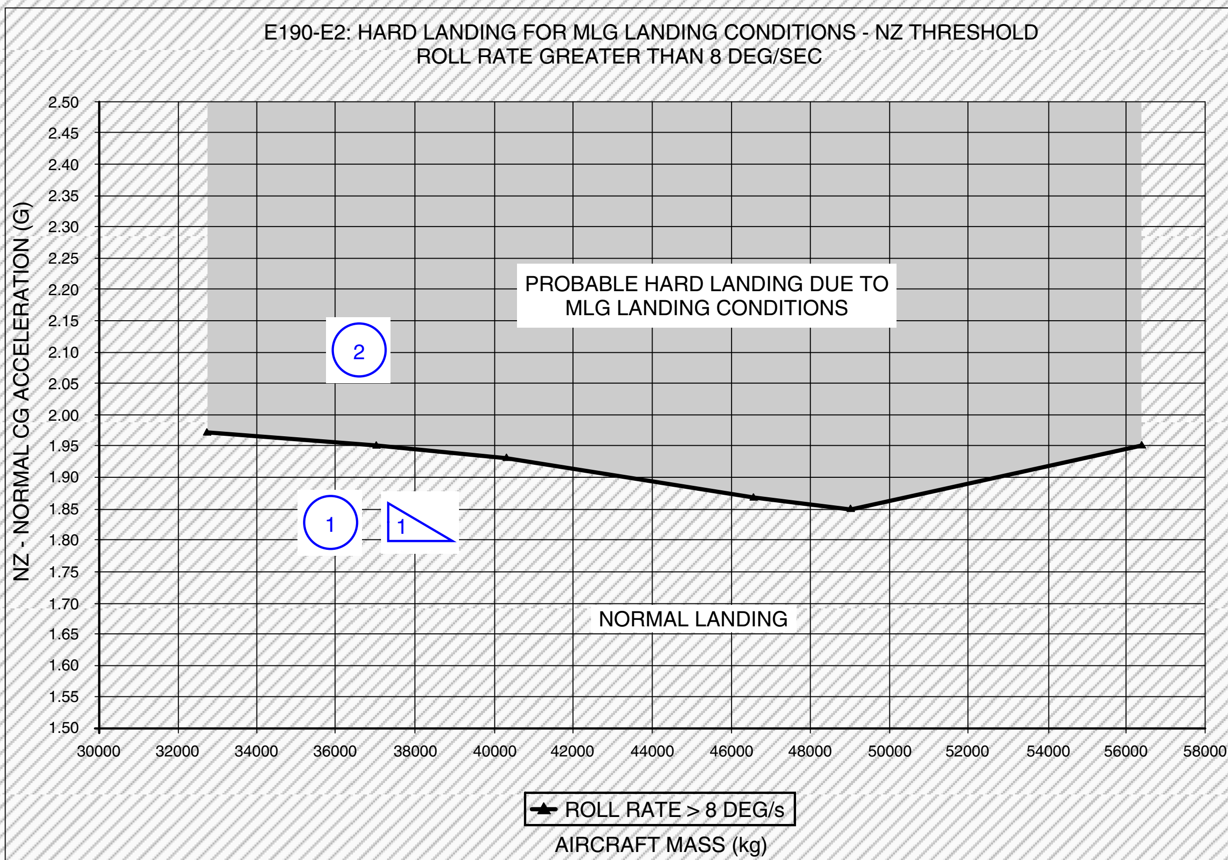
1 PUT THE AIRCRAFT BACK TO SERVICE.

2 DO THE PHASE - II INSPECTION WITHIN 10 FC AFTER PHASE I INSPECTION.

1 PROVIDED THAT NO DAMAGE IS FOUND DURING PHASE I INSPECTION IN CASE OF CREW REPORT.

EM170E2MPP-05-00063G.IDR

EFFECTIVITY: EMBRAER 190-E2 ACFT
Roll rate > 8 DEG/S.
Figure 606



	NORMAL ACCELERATION (NZ) THRESHOLD (G)
AIRCRAFT MASS (kg)	ROLL RATE > 8 DEG/s
32700	1.97
37200	1.95
40300	1.93
46700	1.86
49050	1.85
56400	1.95

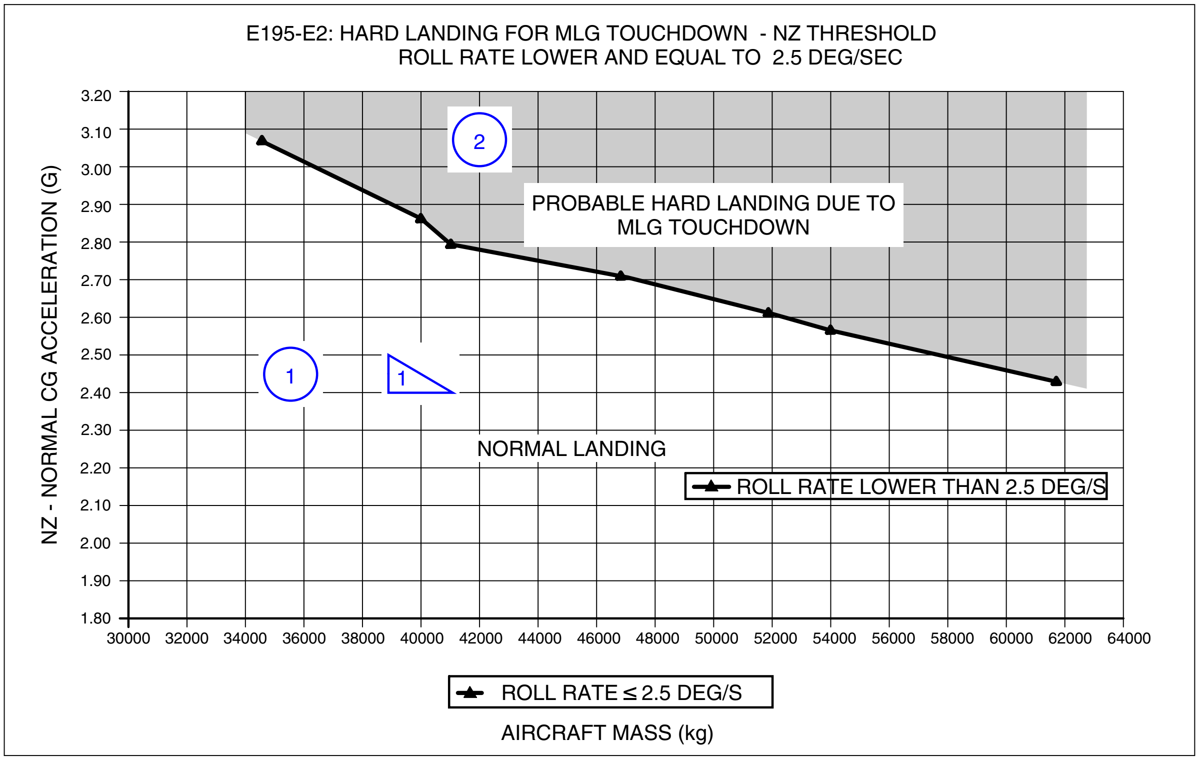
1 PUT THE AIRCRAFT BACK TO SERVICE.

2 DO THE PHASE - II INSPECTION WITHIN 10 FC AFTER PHASE I INSPECTION.

1 PROVIDED THAT NO DAMAGE IS FOUND DURING PHASE I INSPECTION IN CASE OF CREW REPORT.

EM170E2MPP-05-00009E.IDR

EFFECTIVITY: EMBRAER 195-E2 ACFT
Roll rate <= 2.5 DEG/S
Figure 607



	NORMAL ACCELERATION (NZ) THRESHOLD (g)
AIRCRAFT MASS (kg)	ROLL RATE ≤ 2.5 DEG/S
34700	3.05
40000	2.86
40900	2.80
47000	2.71
51850	2.61
54000	2.57
61500	2.43

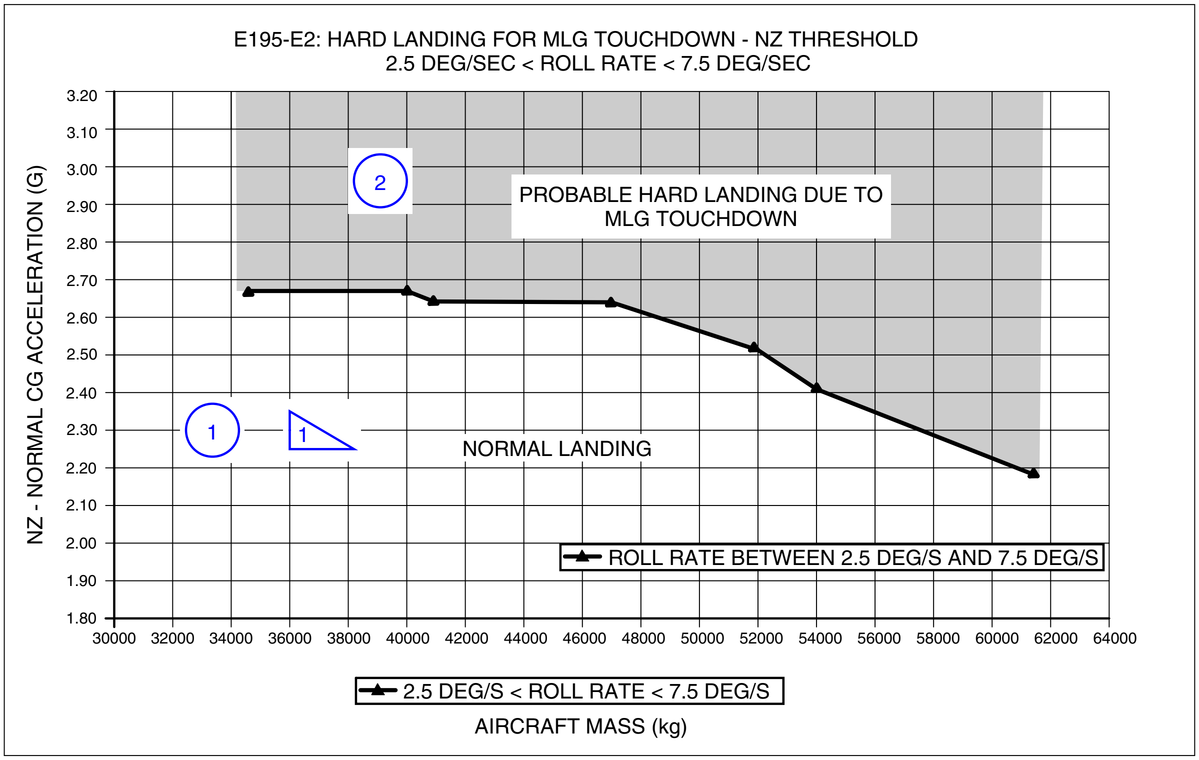
1 PUT THE AIRCRAFT BACK TO SERVICE.

2 DO THE PHASE - II INSPECTION WITHIN 10 FC.

1 PROVIDED THAT NO DAMAGE IS FOUND DURING PHASE I INSPECTION IN CASE OF CREW REPORT.

PV170E2MPP-05-00272B.IDR

EFFECTIVITY: EMBRAER 195-E2 ACFT
2.5 DEG/S < Roll rate < 7.5 DEG/S
Figure 608



	NORMAL ACCELERATION (NZ) THRESHOLD (g)
AIRCRAFT MASS (kg)	2.5 DEG/S < ROLL RATE < 7.5 DEG/S
34700	2.65
40000	2.65
40900	2.65
47000	2.65
51850	2.52
54000	2.42
61500	2.19

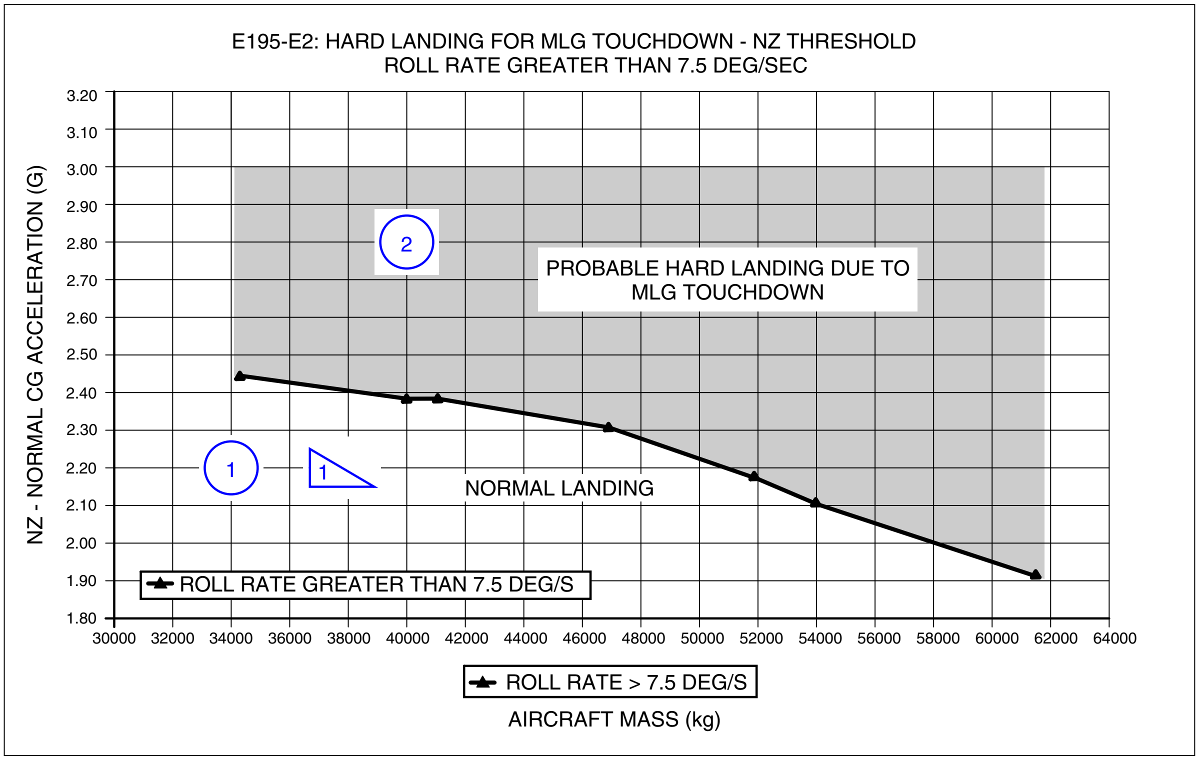
1 PUT THE AIRCRAFT BACK TO SERVICE.

2 DO THE PHASE - II INSPECTION WITHIN 10 FC.

1 PROVIDED THAT NO DAMAGE IS FOUND DURING PHASE I INSPECTION IN CASE OF CREW REPORT.

PV170E2MPP-05-00273B.IDR

EFFECTIVITY: EMBRAER 195-E2 ACFT
Roll rate > 7.5 DEG/S
Figure 609



	NORMAL ACCELERATION (NZ) THRESHOLD (g)
AIRCRAFT MASS [kg]	ROLL RATE > 7.5 DEG/S
34700	2.44
40000	2.38
40900	2.38
47000	2.31
51850	2.17
54000	2.10
61500	1.91

1 PUT THE AIRCRAFT BACK TO SERVICE.

2 DO THE PHASE - II INSPECTION WITHIN 10 FC.

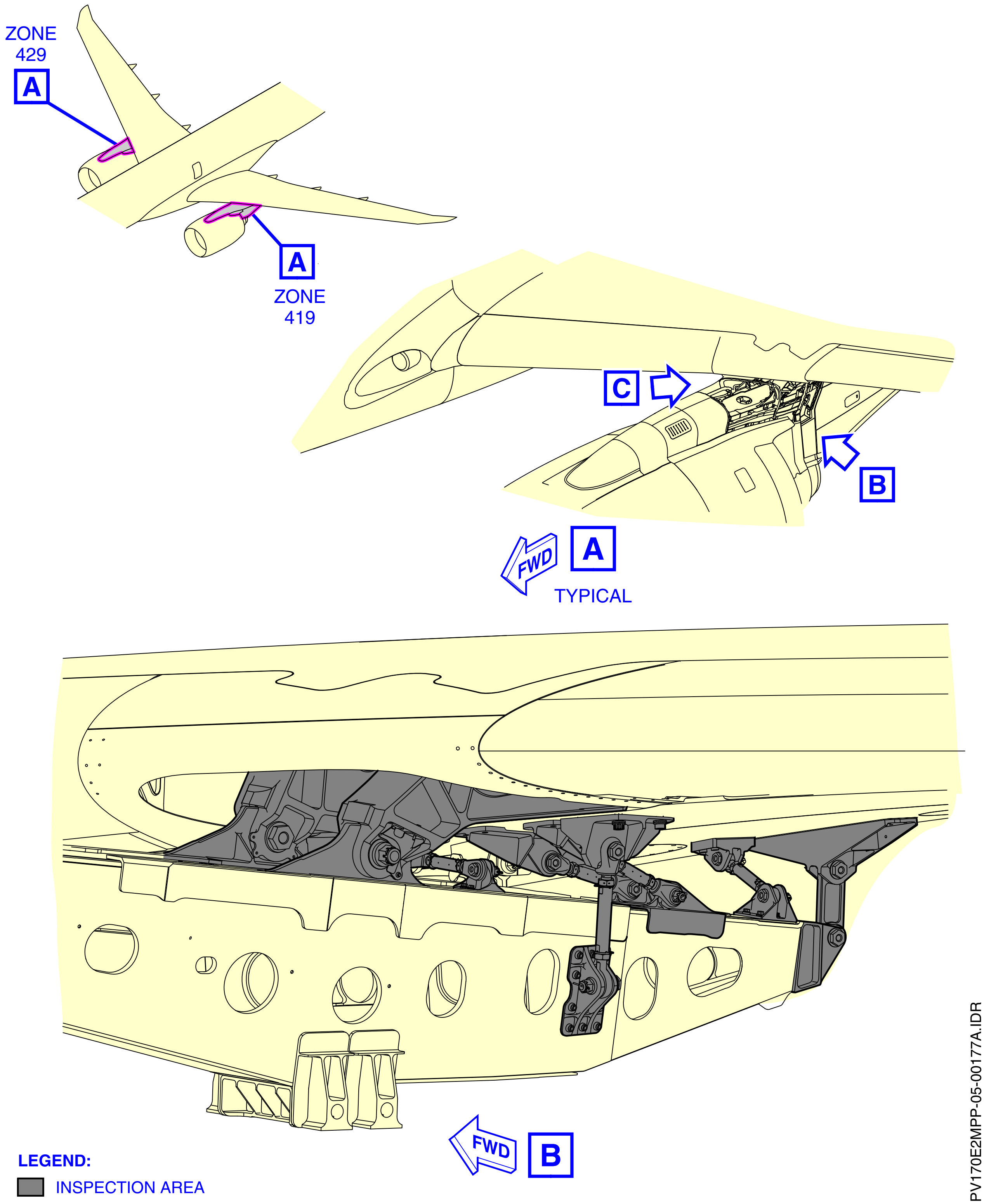
1 PROVIDED THAT NO DAMAGE IS FOUND DURING PHASE I INSPECTION IN CASE OF CREW REPORT.

PV170E2MPP-05-00274B.IDR

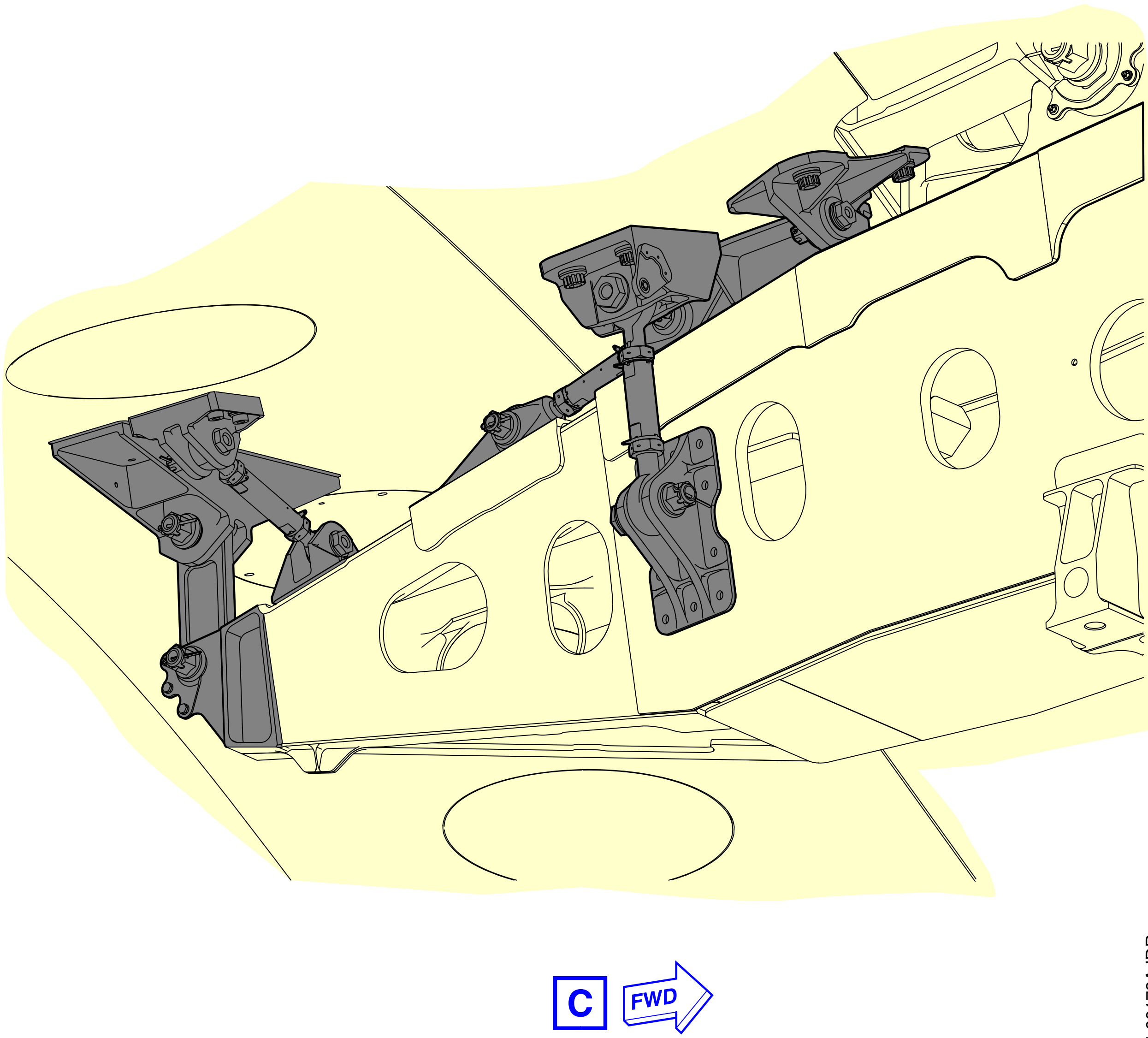
EFFECTIVITY: ALL

Pylon to Wing Inspection

Figure 610 - Sheet 1



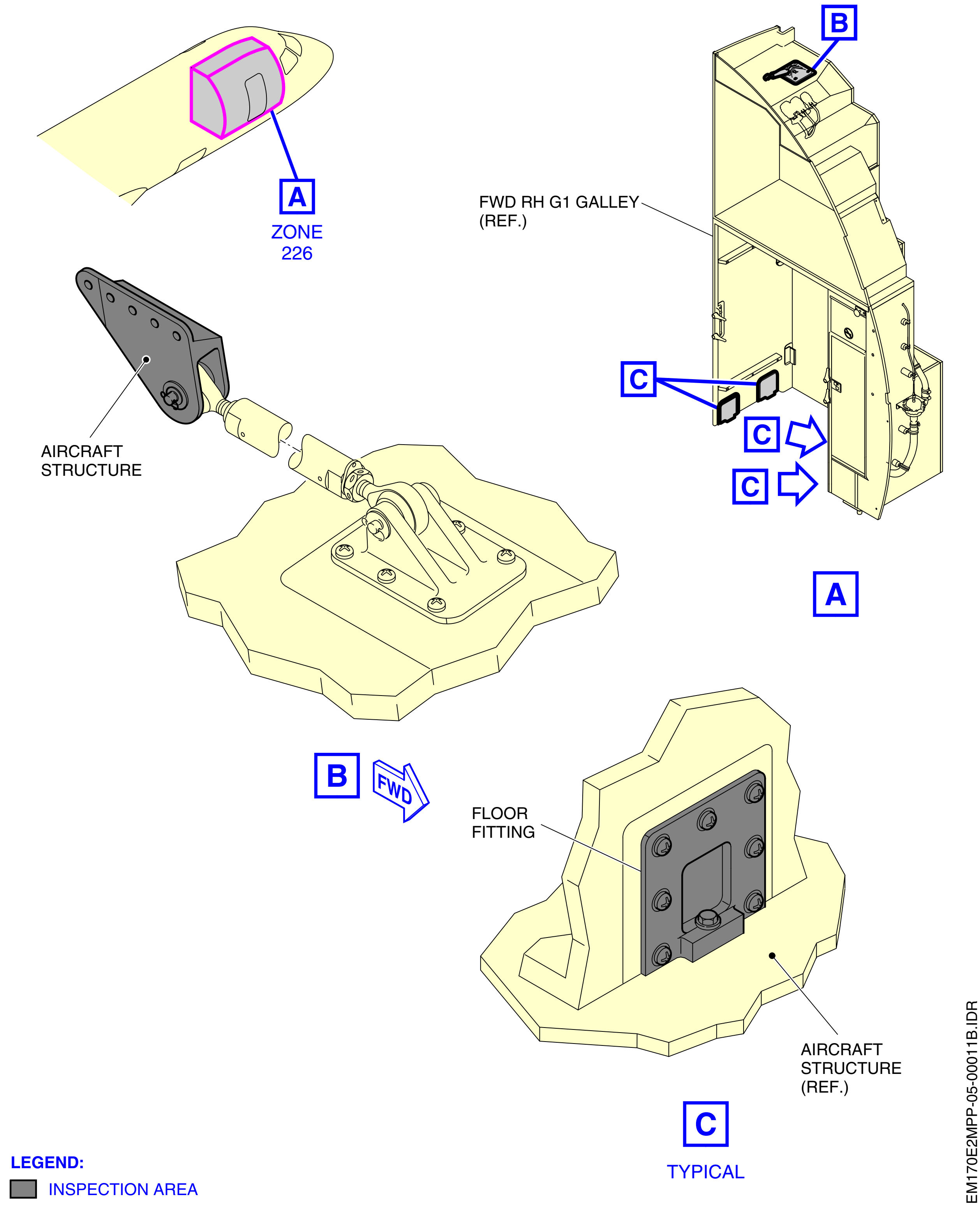
EFFECTIVITY: ALL
Pylon to Wing Inspection
Figure 610 - Sheet 2



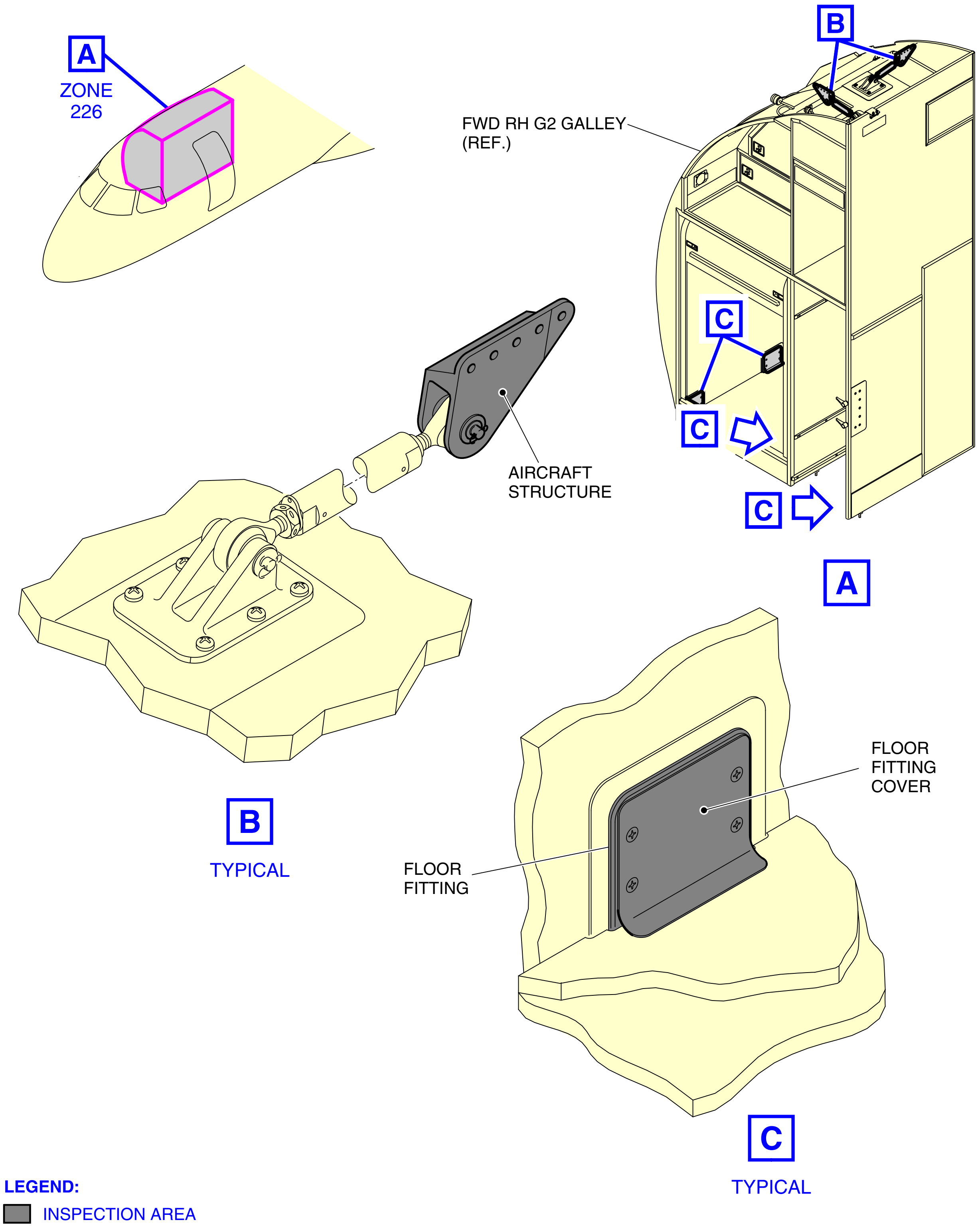
LEGEND:
INSPECTION AREA

PV170E2MPP-05-00178A.IDR

EFFECTIVITY: ALL
Galley Attachments - Inspection
Figure 611 - Sheet 1

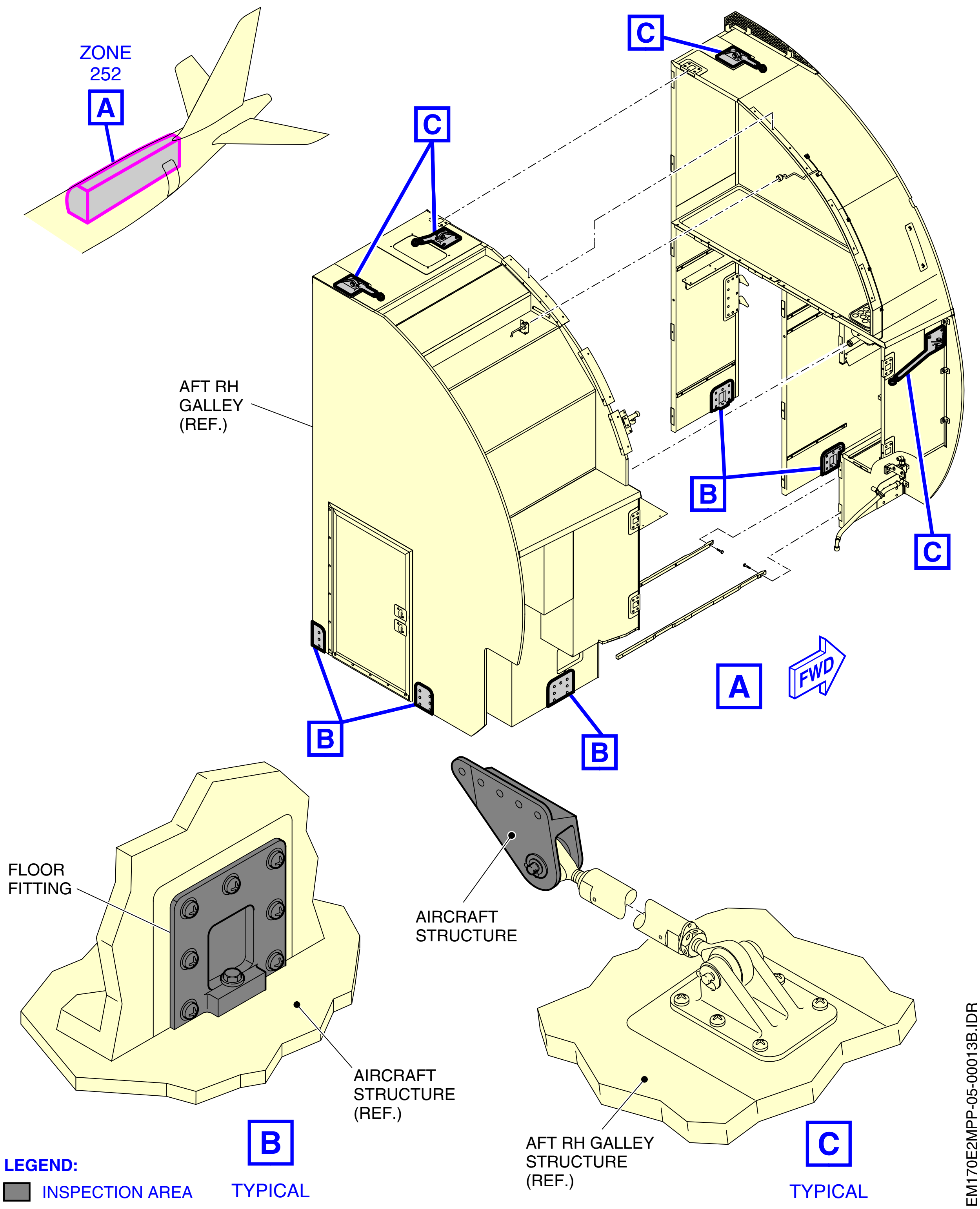


EFFECTIVITY: ALL
Galley Attachments - Inspection
Figure 611 - Sheet 2

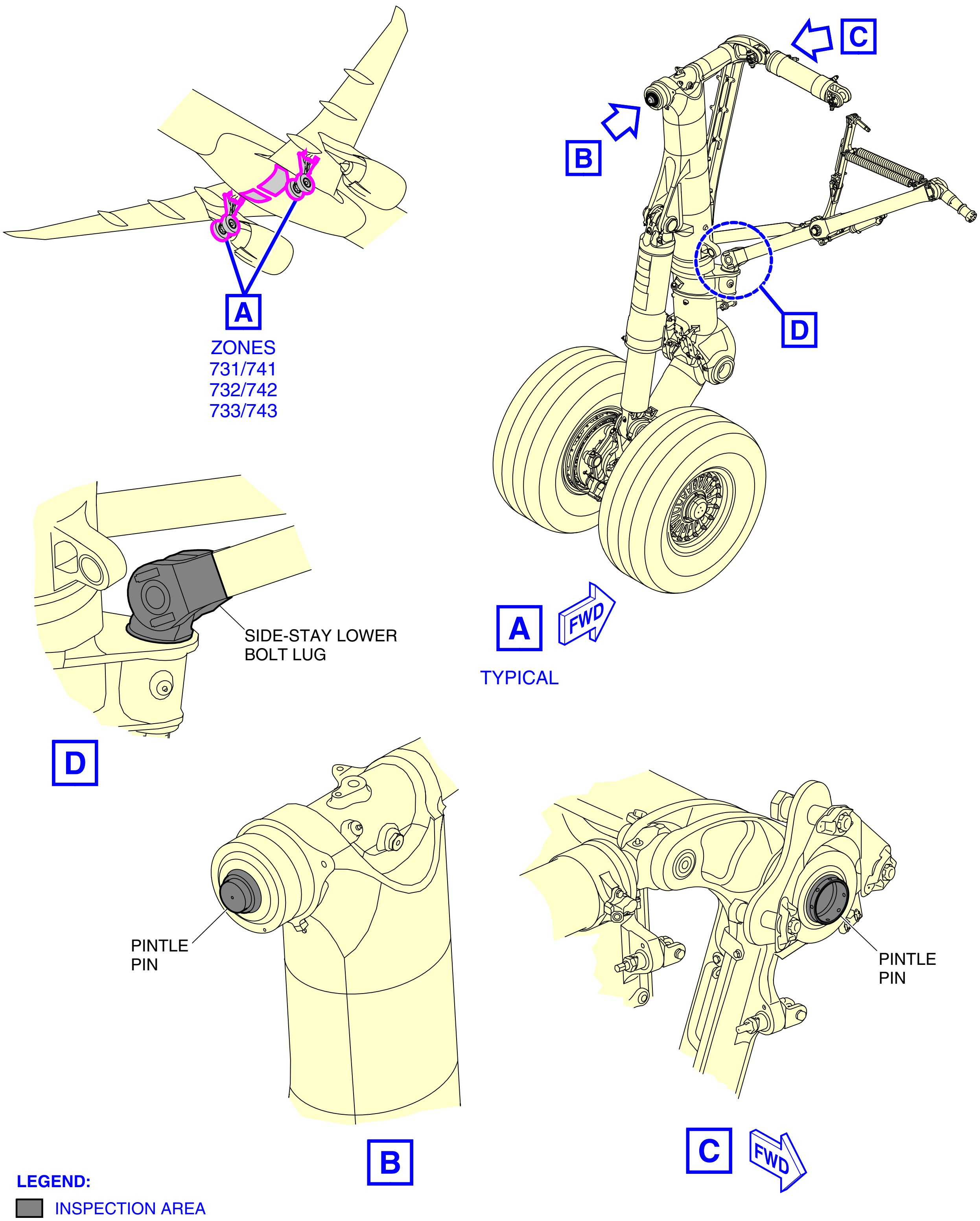


EM170E2MPP-05-00012B.IDR

EFFECTIVITY: ALL
Galley Attachments - Inspection
Figure 611 - Sheet 3

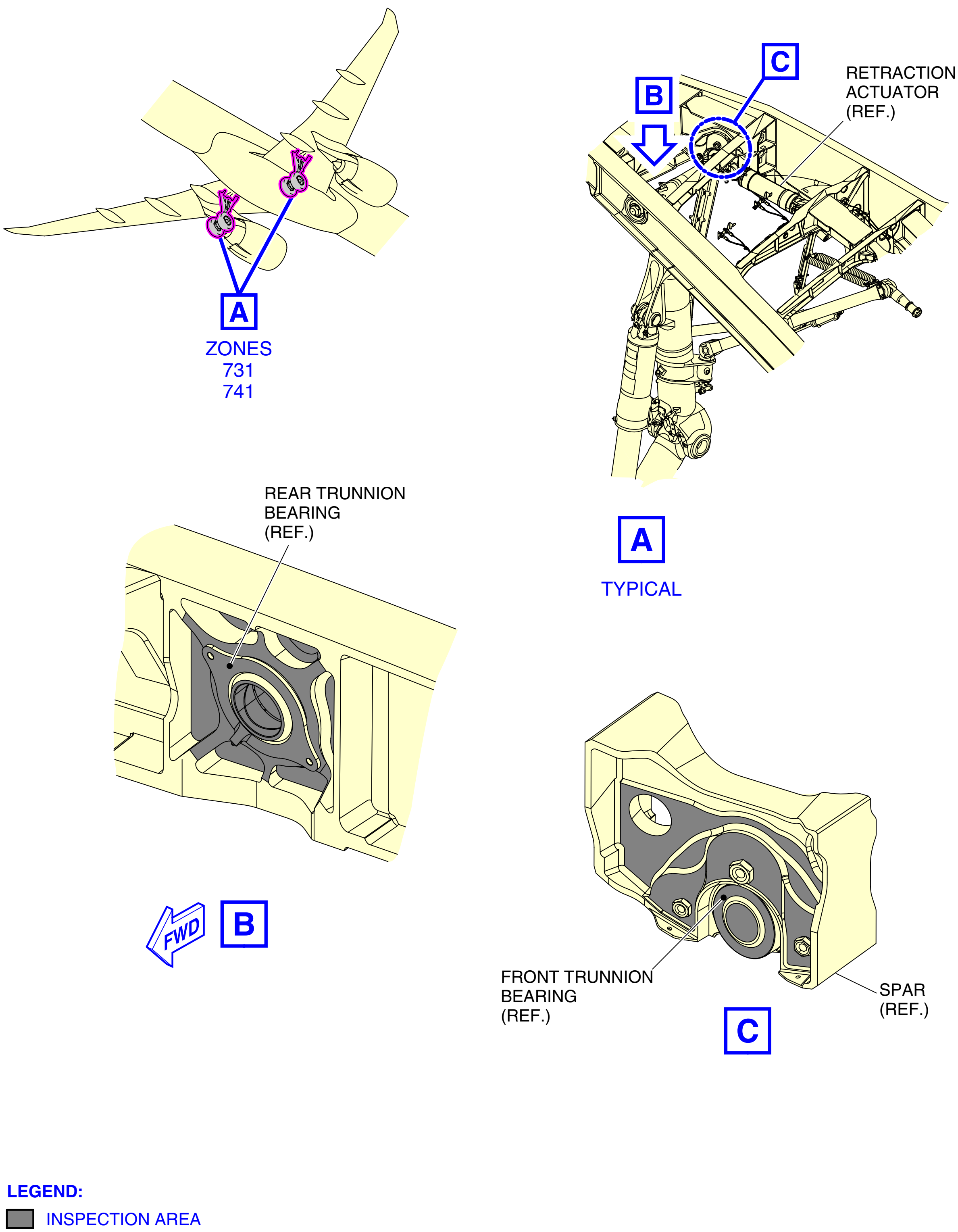


EFFECTIVITY: ALL
MLG Identification and Structures
Figure 612 - Sheet 1

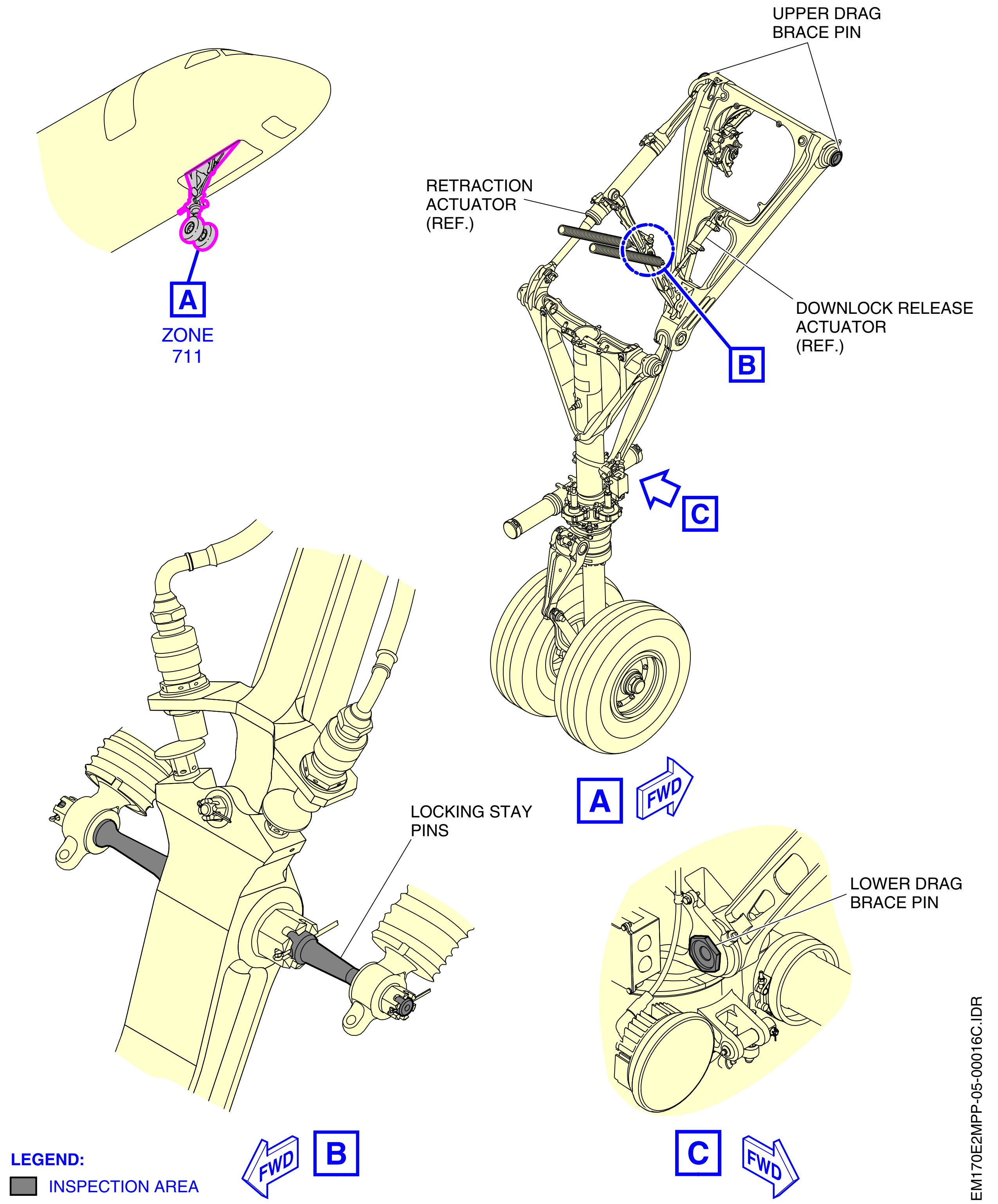


EM170E2MPP-05-00014D.IDR

EFFECTIVITY: ALL
MLG Identification and Structures
Figure 612 - Sheet 2



EFFECTIVITY: ALL
NLG Pins Identification and Structures
Figure 613



EFFECTIVITY: ALL
NLG - phase II Inspection
Figure 614